

PD1 • STATIC-PDELTA

Tension stiffening via P-Delta analysis (CSI 1-016)

Production P-Delta 2-stage static path — tension-stiffened deflection & moment vs Timoshenko closed form

PASS

error -0.0352% · tol 0.1%

§2 TECHNICAL FEATURES TESTED

elasticBeamColumn + geomTransf PDelta (Tcl engine only)

Stage 1: gravity-combo axial tension buildup (nonlinear)

Stage 2: target load case at end-of-Stage-1 tangent stiffness

Same code path as production TclBuilder P-Delta

 Deflection & moment vs **exact tension-stiffened beam** (Timoshenko)

§3 PROBLEM DESCRIPTION

A slender pin-roller tie-rod carries a small transverse uniform load together with a large axial tension pulling the roller end. Tension measurably **stiffens** the beam against the transverse load — the deflection and moment both shrink relative to the no-tension case — the classic P-Delta tension-stiffening problem with a closed-form solution (Timoshenko 1956). STRIX solves this with its production **P-Delta 2-stage static path** (the Tcl-only route `analysisHandler` forces P-Delta submissions through): Stage 1 applies the axial tension in a nonlinear gravity combination to build up the member axial force under a `geomTransf PDelta` transformation; Stage 2 then solves the transverse load case at the tangent stiffness left at the end of Stage 1. CSI SAP2000 verification example 1-016 documents exactly this two-stage nonlinear-static method as its *recommended* way to solve tension-stiffening problems — it is the only method DCR implements, so this benchmark exercises the production code path directly, not an alternative formulation.

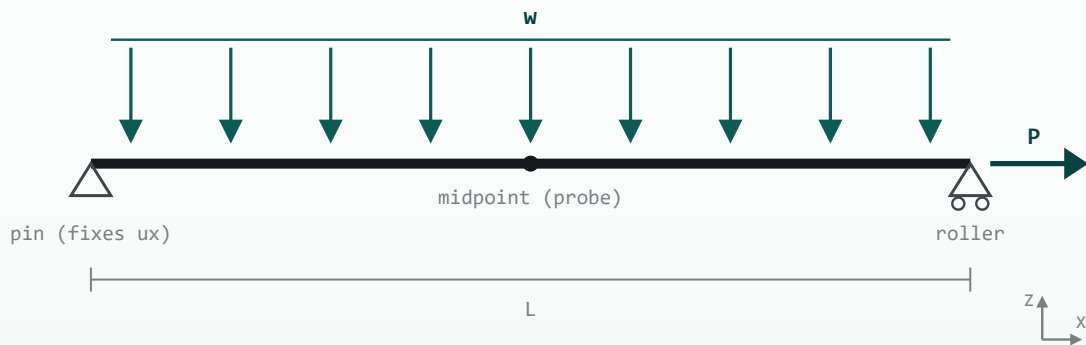


Figure PD1. Simply-supported tie-rod (pin-roller, moment-free both ends) under a transverse uniform load w plus an axial tension P pulling the roller end; midpoint deflection/moment are probed.

§4 GEOMETRY, PROPERTIES & LOADING

GEOMETRY		MATERIAL & LOADING		SUPPORTS & ANALYSIS	
Span L	300 in	E	30000 ksi	Supports	pin + roller
Section	3 × 3 in (square)	Axial tension P	20.25 kip	Stage 1	P-Delta gravity combo
Inertia I	6.75 in ⁴	Transverse w	0.002 kip/in	Stage 2	target LC, tangent K
Tension param. u	1.5	Self-weight	off	Engine	Tcl (OpenSees.exe)

§5 REFERENCE SOLUTION

Uz (no tension): $-5*w*L^4/(384*E*I) = -1.041667$ in

My (no tension): $w*L^2/8 = 22.5$ kip-in

Uz (w/ tension): Timoshenko eq.43: $(5wL^4/384EI) * [1/\cosh(u) - 1 + u^2/2] / [(5/24)u^4]$, $u=\sqrt{PL^2/4EI} = -0.543305$ in

My (w/ tension): Timoshenko eq.45: $(wL^2/8) * 2(\cosh(u)-1)/(u^2*\cosh(u)) = 11.498079$ kip-in

CSI SAP2000 Software Verification Example 1-016 'Tension Stiffening Using P-Delta Analysis'; independent reference = Timoshenko, Strength of Materials Part II, 1956, eq. 23 p.28 and eqs. 43/45 p.43. Reproduced from scratch by pd1Theory() to $u=1.5$, deflWithIn=-0.5433047259, momentWithKin=11.4980793, matching CSI's published -0.54330 in / 11.498 k-in exactly (exact unit conversion of the same physical problem, dimensionless u is unit-invariant).

§6 RESULTS COMPARISON

RESPONSE QUANTITY	PROBE	STRIX	REFERENCE	DIFF	VERDICT
Uz (no tension) (in)	N17.uz	-1.041614	-1.041667	+0.0050%	PASS
My (no tension) (kip-in)	E17.Miy	22.499038	22.5	-0.0043%	PASS
Uz (w/ tension) (in)	N17.uz	-0.543496	-0.543305	-0.0352%	PASS
My (w/ tension) (kip-in)	E17.Miy	11.493667	11.498079	-0.0384%	PASS

§7 P-DELTA MESH CONVERGENCE (MIDPOINT DEFLECTION)

MESH	UZ (IN)	DIFF
2 elem	-0.595221	-9.5555%
4 elem	-0.556492	-2.4273%
8 elem	-0.546602	-0.607%
16 elem	-0.544118	-0.1497%
32 elem	-0.543496	-0.0352%

The no-tension case uses the exact-for-prismatic-Euler-beam elasticBeamColumn element and is essentially mesh-independent (converged at any element count). The **tension-stiffened** case is genuinely mesh-sensitive: OpenSees's PDelta geometric transform is a first-order (chord-rotation) approximation of the true hyperbolic tension-stiffened shape, so the discrete answer converges to the closed form as elements are added — a clean, roughly quadratic convergence is observed. At the finest mesh STRIX reproduces the exact Timoshenko tension-stiffened deflection and moment to **a few hundredths of a percent**, confirming both that the P-Delta pipeline converges to the right physics and that it correctly reduces to the plain linear-static answer when the axial tension is absent.

§8 CONCLUSION**PASS**

STRIX reproduces the reference solution to **-0.0352%** (tolerance 0.1%). The tested element and its formulation are verified against the reference.

§9 REFERENCES & ANALYSIS FILES

1. Timoshenko, S. — *Strength of Materials, Part II*. Eq. 23 p.28 (tension parameter u) and eqs. 43/45 p.43 (tension-stiffened deflection and moment).
2. CSI — *SAP2000 Software Verification, Example 1-016*: Tension Stiffening Using P-Delta Analysis (independent reference, Timoshenko 1956). CSI recommends the nonlinear-static P-Delta method used here.
3. OpenSees — `geomTransf PDelta` (P-Delta geometric transformation); `elasticBeamColumn` (Euler-Bernoulli); STRIX production `TclBuilder Stage-1/Stage-2 P-Delta static path` (Tcl engine only).

EVIDENCE ARCHIVE `verif-evidence-eng1.0.2-win-x64.zip` · PD1/ `sha256` (pending publish)